

Maximum Mode Timing Diagrams

Q. Discuss the Memory Read and Write timing diagrams for the 8086 microprocessor operating in Maximum Mode. How does it differ from Minimum Mode

[DEF] Definition to Remember

In Maximum Mode, the 8086 delegates control signal generation to the 8288 Bus Controller. Instead of generating RD' or WR', the CPU outputs early status codes (S2', S1', S0') just before T1. The 8288 decodes these to issue the correct timing signals.

1. The Role of Status Signals

* Pre-T1 Phase: Before the T1 clock cycle even begins, the 8086 updates S2', S1', S0' to indicate the upcoming cycle (e.g., 101 for Memory Read, 110 for Memory Write).

2. Memory Read Timing (Maximum Mode)

* Pre-T1: S2', S1', S0' = 1, 0, 1 (Memory Read).

3. Memory Write Timing (Maximum Mode)

* Pre-T1: S2', S1', S0' = 1, 1, 0 (Memory Write).

4. Key Differences from Minimum Mode

* Signal Source: 8288 generates ALE, DEN, DT/R', not the CPU.

[MUST-WRITE] Must-Write Points (for 10 marks)

1. In Max Mode, the 8288 Bus Controller handles the timing signals.
2. CPU communicates via S2', S1', S0' status lines, which change *before* T1 begins.

3. T1: 8288 generates ALE to latch the address put out by the CPU.
4. T2: 8288 asserts specific commands like MRDC' (Memory Read) or MWTC' (Memory Write) instead of generic RD/WR.
5. DEN is generated by the 8288 and is active-high in Max Mode.
6. Advanced Write Commands (AMWC') can be used to give slow memory more time to write.
7. In T3, the status lines revert to 1, 1, 1 (Idle) to signal the end of the operation.

[RECALL] Quick Recall

Max Mode Timing -> 8288 in charge -> S2,S1,S0 set before T1 -> 8288 generates ALE (T1), MRDC/MWTC (T2), DEN (T2) -> Status goes idle (1,1,1) in T3 -> DEN is active-high